TPVA-22: Define Feasibility Rules and Validation Statuses

Feasibility & Ranking Engine — Member 4

Owner: M4 (SAND)

Target: 29 October 2026 | Status: Draft for Review

Contents

1 Objective 1

2 Validation Statuses 2

3 Constraint Classification 2

4 Validation Rules Table 2

4.1 R01 — Budget Validation . . . 2
4.2 R02 — Trip Duration Validation . . . 3
4.3 R03 — Schedule Conflict Validation . . . 3
4.4 R04 — Opening-Hour Validation . . . 3
4.5 R05 — Inter-stop Travel-Time Validation . . . 3
4.6 R06 — Daily Travel-Distance Validation . . . 3
4.7 R07 — Additional User Mandatory Constraints . . . 3
4.8 R08 — Preferences . . . 3

5 Overall Status Aggregation 4

6 Initial Validation Test Scenarios 4

7 Proposed Validation Output Contract 5

8 Module Integration Responsibilities 5

9 Jira Acceptance Checklist and Open Decisions 5

10 Source and Scope Note 6

1 Objective

Define deterministic validation rules for evaluating travel itineraries generated by M3. The M4 validator must:

- Distinguish mandatory constraints from optional preferences.
- Return exactly one overall status: `FEASIBLE`, `INFEASIBLE`, or `UNVERIFIED`.
- Identify budget, duration, schedule, opening-hour, and applicable travel-limit problems.
- Explain confirmed violations and disclose missing essential data.
- Evaluate plans independently of claims generated by an AI model.
- Produce findings that can be consumed by the ranking engine and frontend.

2 Validation Statuses

Status	Definition	Ranking
`FEASIBLE`	All applicable mandatory constraints are verified and satisfied.	Eligible
`INFEASIBLE`	At least one mandatory constraint is confirmed violated.	Ineligible
`UNVERIFIED`	No confirmed mandatory violation exists, but essential information is insufficient to verify every applicable mandatory constraint.	Ineligible

Principle: Missing information is not proof of failure. Unknown opening hours cannot be assumed to mean that a destination is closed.

3 Constraint Classification

Type	Definition	Example	Treatment
System mandatory	Conditions required for an executable itinerary	No overlapping activities	Validation
User mandatory	Explicit non-negotiable limits set by the user	Maximum 3,000,000 VND	Validation
User preference	Flexible desirable properties	Prefer nature attractions	Ranking

The same attribute can be either a mandatory constraint or a preference depending on the request. For example, "no more than 40 km per day" is a hard limit, while "prefer shorter trips" is a preference.

4 Validation Rules Table

ID	Rule	Validation condition	Failure / uncertainty
R01	Budget limit	Verified known cost must respect hard budget; assess missing essential costs	Confirmed excess: `INFEASIBLE`; unresolved essential cost: potentially `UNVERIFIED`
R02	Trip duration	Planned activities fit required dates and duration	Outside permissible period: `INFEASIBLE`
R03	Schedule conflict	Activity time intervals must not overlap	Confirmed overlap: `INFEASIBLE`
R04	Opening hours	Activity fits within a known applicable opening window	Known closure: `INFEASIBLE`; essential hours unknown: `UNVERIFIED`
R05	Travel time	Gap between consecutive activities is sufficient for travel	Too little time: `INFEASIBLE`; essential leg time unknown: `UNVERIFIED`
R06	Travel distance	Sum of travel legs respects any mandatory distance limit	Limit exceeded: `INFEASIBLE`; relevant legs missing: `UNVERIFIED`
R07	User mandatory limits	All supported explicit hard requirements are met	Confirmed violation: `INFEASIBLE`
R08	User preferences	Preferences are evaluated separately from hard feasibility	Lower preference score, not automatic infeasibility

4.1 R01 — Budget Validation

For known itemized amounts, the reference model is:

$$C_{\text{total}} = C_{\text{transport}} + C_{\text{accommodation}} + C_{\text{dining}} + C_{\text{activities}}. \tag{1}$$

A known cost exceeding a hard cap is a confirmed violation, even if additional costs remain unknown. If essential amounts are missing and no violation is established, the plan cannot be confirmed feasible. Costs and budget must use comparable currency, party-size, and trip-level units.

4.2 R02 — Trip Duration Validation

Scheduled activity dates and trip duration must match the mandatory period in the normalized `TravelRequest`. Date fields must be valid, and the planned activities must fall within that period.

4.3 R03 — Schedule Conflict Validation

For valid, same-day activity intervals $A = [s_A, e_A)$ and $B = [s_B, e_B)$, overlap exists if

$$s_A < e_B \quad \wedge \quad s_B < e_A. \tag{2}$$

Using half-open intervals means one activity ending exactly when another starts is not an overlap. Travel-time feasibility must still be checked independently. Cross-midnight activity intervals require a consistent date-time representation.

4.4 R04 — Opening-Hour Validation

For an applicable known opening window, the planned visit must fit within that window. Unknown essential opening hours are not interpreted as a confirmed closure. Multiple windows, holidays, and exceptions should be represented explicitly when supported by the dataset.

4.5 R05 — Inter-stop Travel-Time Validation

For consecutive activities A and B with a known travel-time estimate:

$$e_A + t(A, B) \leq s_B. \tag{3}$$

If the necessary travel-time estimate is unknown, the validator records an unresolved essential check rather than assuming zero.

4.6 R06 — Daily Travel-Distance Validation

For consecutive itinerary stops $P_1, \ldots, P_n$, daily distance is

$$D_{\text{day}} = \sum_{i=1}^{n-1} d(P_i, P_{i+1}). \tag{4}$$

Compare the sum with a mandatory daily limit when one was specified. Missing applicable legs must be disclosed; absent a mandatory distance limit, distance may still contribute to ranking.

4.7 R07 — Additional User Mandatory Constraints

Evaluate explicit hard limits that the system supports and that M1 has normalized. Ambiguous and unsupported requirements need a recorded response or clarification; they must not be silently treated as satisfied.

4.8 R08 — Preferences

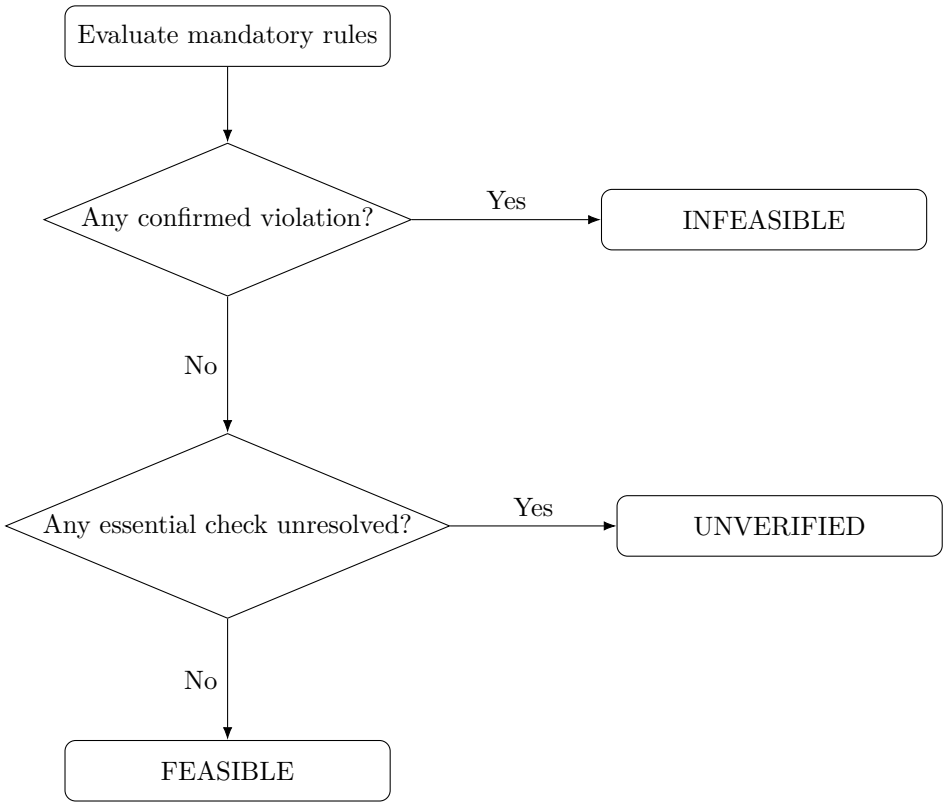
Preferences (e.g., nature activities, lower costs, convenient routes) influence ranking but do not override mandatory feasibility checks. Ranking is performed only for verified feasible plans.

5 Overall Status Aggregation

Let S_i be the result of each applicable mandatory validation rule. Overall evaluation follows this precedence:

1. If *any* mandatory rule confirms a violation, return `INFEASIBLE`.
2. Otherwise, if *any* essential mandatory check cannot be resolved, return `UNVERIFIED`.
3. Otherwise, return `FEASIBLE`.

The sequence `INFEASIBLE` > `UNVERIFIED` > `FEASIBLE` denotes *decision precedence*, not a numeric score. The engine should collect all detectable issues instead of halting at the first failure.



6 Initial Validation Test Scenarios

Assume all mandatory checks not mentioned in a scenario are satisfied.

Test ID	**Scenario**	**Expected**
TC01	All mandatory constraints satisfied	`FEASIBLE`
TC02	Known total exceeds hard budget	`INFEASIBLE`
TC03	Essential accommodation cost unknown	`UNVERIFIED`
TC04	Two scheduled activities overlap	`INFEASIBLE`
TC05	Activities meet at an endpoint and no travel is required	`FEASIBLE`
TC06	Activity after confirmed closing time	`INFEASIBLE`
TC07	Essential opening hours unavailable	`UNVERIFIED`
TC08	Travel requires 30 minutes; only 10 minutes are available	`INFEASIBLE`

Test ID	Scenario	Expected
TC09	Daily distance exceeds an explicit hard limit	`INFEASIBLE`
TC10	Preference mismatch, all hard checks satisfied	`FEASIBLE`
TC11	Budget violation and unknown opening hours	`INFEASIBLE`
TC12	Essential leg travel time unknown, no confirmed violation	`UNVERIFIED`

These are proposed test specifications, **not** evidence that implementation tests have passed.

7 Proposed Validation Output Contract

The following JSON is an illustrative response to be reviewed with M1, M2, M3, and M6. Field names are not yet finalized.

```
{
  "itineraryId": "PLAN_001",
  "status": "INFEASIBLE",
  "eligibleForRanking": false,
  "issues": [
    {
      "ruleId": "R01",
      "status": "INFEASIBLE",
      "reason": "Confirmed cost exceeds budget",
      "details": {
        "budget": 3000000,
        "knownCost": 3500000,
        "currency": "VND"
      }
    },
    {
      "ruleId": "R04",
      "status": "UNVERIFIED",
      "reason": "Opening hours unavailable",
      "activityId": "ACT_003"
    }
  ]
}
```

A single overall itinerary status must be kept separate from individual validation findings.

8 Module Integration Responsibilities

Module	Information exchanged with M4
M1	Normalized mandatory constraints and preferences
M2	Cost data, opening hours, travel times, and distances
M3	Generated or revised itinerary
M5	Status, failure reasons, and missing-data warnings
M6	Shared contracts, integration, and test evidence

9 Jira Acceptance Checklist and Open Decisions

- ☐ Document three statuses and the precedence rules.

- ☐ Document mandatory constraints versus preferences.
- ☐ Define and illustrate validation rules R01–R08.
- ☐ Record initial test cases and expected outcomes.
- ☐ Review the requirements and schemas with relevant module owners.
- ☐ Implement/review agreed code and attach test or pull-request evidence.
- ☐ Update Jira and obtain approval before marking the task Done.

Decisions requiring team confirmation:

1. Whether the budget is per person or per group, and whether any flexibility is explicitly supported.
2. Which missing fields are essential enough to prevent confirmed feasibility.
3. Which additional user-defined hard constraints the MVP supports and how unsupported requests are handled.
4. Shared `TravelRequest` and `Itinerary` field names, units, and missing-value representations.

Current completion status: Design draft ready for team review. Integration and executed test evidence remain pending. This document alone does not establish Jira acceptance.

10 Source and Scope Note

The M4 scope and acceptance expectations are based on the team's *Personalized Travel Planning Virtual Assistant* proposal, especially Section 6.4. The rule IDs, detailed conditions, tests, and response format above are implementation proposals developed for TPVA-22 and require team review.